LAB 12

Regime-Filter Stage: PCA and K-Means as a Gate on the Signals

Item	Detail
Topic	Topic 3: Machine Learning Models, Statistical Modelling and A/B Testing for Trading Signals
Pipeline stage	Regime filter
Duration	20 min
Simulator page	http://localhost:8000/#/research/regime
Mode	No coding. Runs ONLINE with your Alpaca paper keys (Integrations page), or OFFLINE on the demo data and the built-in paper broker. Paper trading only.

Lion City Capital and every scenario in these materials are fictitious and for training only. All trading in this course is paper (simulated) trading. Nothing in these materials is investment advice or a recommendation to trade any security.

Scenario

The risk team suspects the desk's signals behave differently in calm and stressed markets. It asks for an unsupervised regime model that can switch signals off in stress - fitted walk-forward so that the gate never uses information from the future.

Objective

Find market regimes with walk-forward K-means and gate a signal in the high-volatility regime.

Before You Start

- Make sure the Trading Simulator is running and open http://localhost:8000 > Course Labs > Lab 12. It is installed once - see the guide 'Download, Install and Start the Trading Simulator Software' in the Learner Guide.
- Record every result you are asked to check - you will use it in the debrief and in the later labs.

Lab Steps - In the Trading Simulator

No coding is needed. Make sure the Trading Simulator is running (installed once - see the guide 'Download, Install and Start the Trading Simulator Software') and open http://localhost:8000. In the sidebar open Course Labs > Lab 12: the app shows the same steps with a Go button that opens each page pre-filled. Every step works offline on the bundled demo data and the built-in paper broker.

Step 1. Analyse regimes

Research > Regimes: SPY, 3 clusters, gate the EMA crossover.

Open: http://localhost:8000/#/research/regime (Research > Regimes SPY)

Check: PC1 explains the largest share of variance (the market factor); the 2020 crash is in the stress regime.

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Step 2. Backtest the regime-filtered strategy

Backtest 'Regime-filtered strategy (K-means gate)'.

Open: http://localhost:8000/#/backtest?strategy=regime_filtered&symbols=SPY (Backtest SPY regime filter)

Check: Compare max drawdown with the ungated EMA crossover.

Expected in the app: PCA explained variance, a regime timeline and gated vs ungated performance.

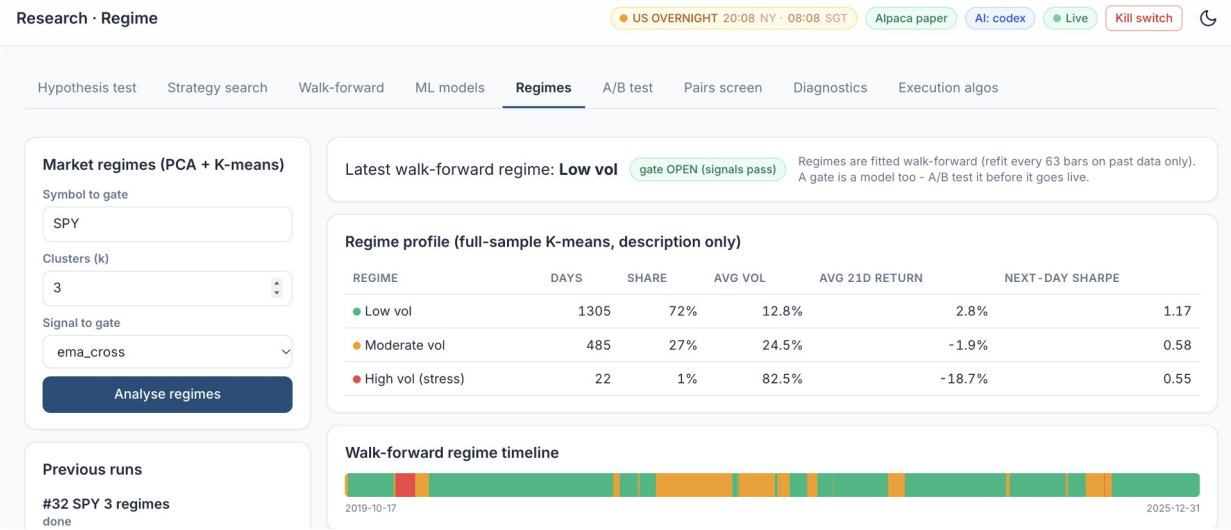


Figure: Lab 12 in the Trading Simulator.

Optional: Let AI Write the Code

This lab needs no coding. To see, run or customise the code behind it, click a link: the AI opens with the prompt filled in and writes the code for you. All prompts are also in labs/Prompt_Library.html and .pdf.

Market regimes with PCA and K-means - run it in Google Colab
I am a beginner and do not write code. Write ONE complete Python script that runs as-is in a Google Colab notebook. Start with a !pip install line for any library Colab lacks, use pandas, numpy, matplotlib, scipy, scikit-learn, statsmodels and yfinance only, get daily prices from yfinance (auto_adjust=True unless I say otherwise), put a one-line comment above each section, print results as clear tables and charts, and after the code explain in 5 short bullet points how to read the output. Paper trading and education only. Task: label SPY market regimes for 2018-2025 with PCA and K-means (3 clusters) on the 21-day return, 21-day volatility and 63-day return, fitting on past data only (refit once a year). Plot the regimes on the price chart, then backtest an EMA 20/50 crossover with and without a gate that stays flat in the high-volatility regime (trade at the next day's open with a 0.05% cost per trade) and compare CAGR, Sharpe and maximum drawdown.

Open in Claude | Open in ChatGPT | Open Google Colab

Debrief

Unsupervised learning finds structure without labels: PCA compresses correlated assets into a few drivers and K-means groups days into regimes. Fitting the regimes walk-forward is part of validating the model, and testing the gate signal by signal after costs shows that a filter can help one strategy and hurt another.

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